

ACTIVE DIRECTORY DEPLOYMENT

AND

WAZUH SIEM INTEGRATION

Domain Controller Setup | Audit Policy | Endpoint Monitoring | Attack Simulation | Detection

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Project Type:	Hands-On Active Directory Security & SIEM Lab
Environment:	Windows Server 2022 (DC) Windows 10 Client Wazuh Manager

This project demonstrates the end-to-end deployment of an Active Directory domain, security auditing configuration, and integration with Wazuh SIEM to detect a simulated backdoor-account attack.

1. Project Overview

This project builds on the earlier hands-on Wazuh SIEM lab (log collection, File Integrity Monitoring, custom rules, and email alerting on a standalone Windows 10 endpoint) and extends it into a more realistic enterprise-style scenario centered on Active Directory.

The goal was to deploy a fully functioning Active Directory domain, join a Windows client to that domain, enable Windows security auditing, integrate the Domain Controller with Wazuh SIEM, and validate the entire pipeline by simulating a backdoor-account attack and confirming that Wazuh correctly detected and reported it.

This project covers:

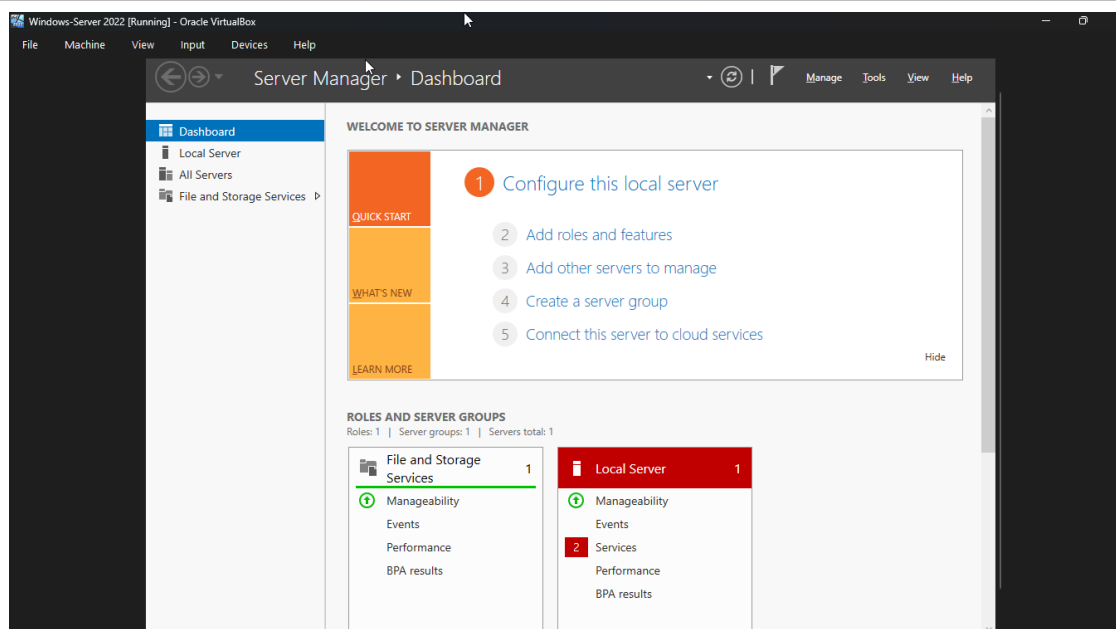
- Deployment and verification of a Windows Server 2022 Domain Controller
- Creation of an Organizational Unit (OU) structure with Users, Groups, Workstations, and Servers
- Domain-joining a Windows 10 client and verifying membership
- Wazuh agent deployment on the Domain Controller
- Configuration of Advanced Windows Audit Policy for AD-relevant events
- Simulation of a backdoor-account attack and detection validation in Wazuh

Tools & Environment

Domain Controller	Windows Server 2022 (hostname: DC01)
Domain	qasimlab.local NetBIOS: QASIMLAB
DC IP Address	192.168.56.110 (static)
Windows Client	Windows 10 (hostname: WIN10-CLIENT01)
SIEM	Wazuh Manager — 192.168.56.118

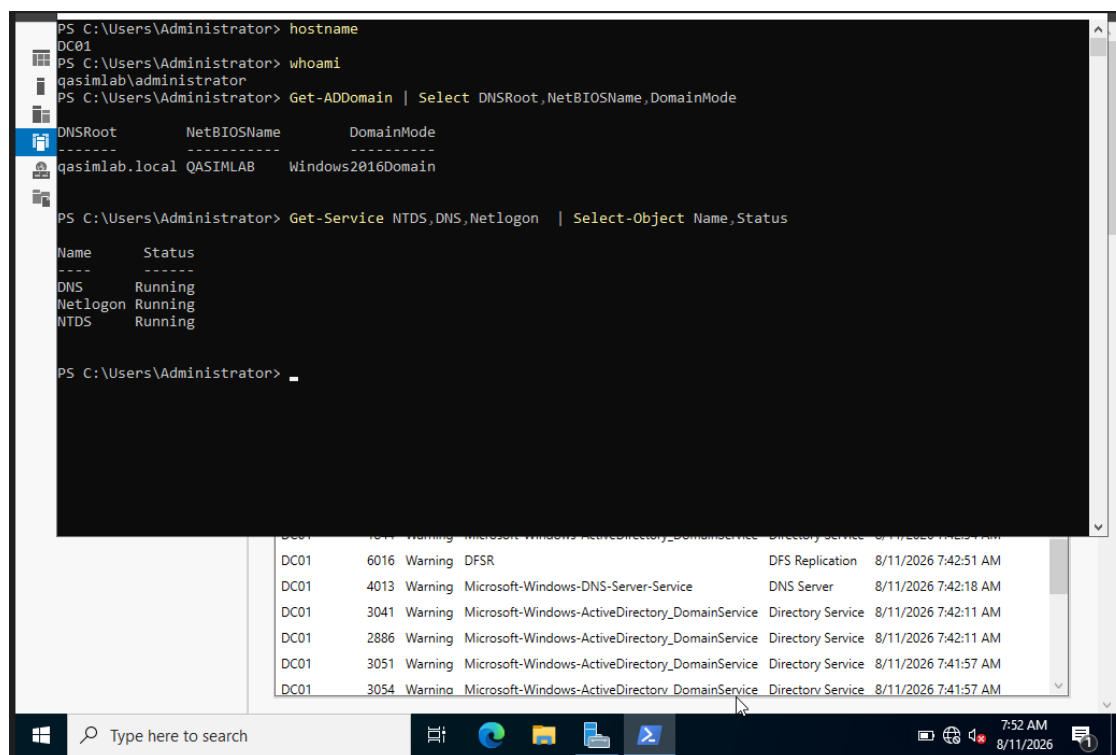
2. Domain Controller Deployment & Verification

A Windows Server 2022 virtual machine was deployed and promoted to a Domain Controller. Active Directory Domain Services (AD DS) and DNS roles were installed, and a new forest was created for the domain qasimlab.local with NetBIOS name QASIMLAB.



Windows Server 2022 running in Oracle VirtualBox — Server Manager dashboard after initial configuration.

After promotion, the domain was verified from PowerShell. The hostname resolved to DC01, and the currently logged-in account was confirmed as qasimlab\administrator. The Get-ADDomain cmdlet confirmed the DNS root (qasimlab.local), NetBIOS name (QASIMLAB), and the domain functional mode (Windows2016Domain). The core AD services — DNS, Netlogon, and NTDS — were all confirmed as Running.

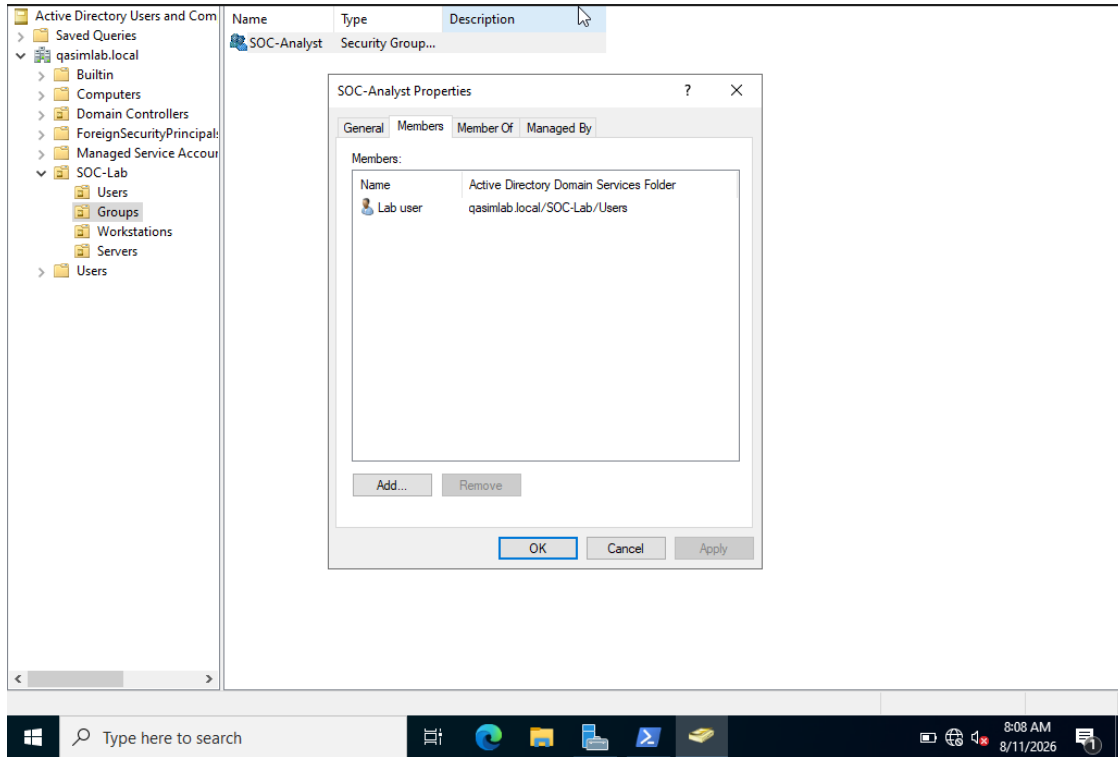


PowerShell verification of DC01: hostname, domain identity, and core AD service status.

3. Organizational Unit, User & Group Structure

To reflect a realistic AD layout, an Organizational Unit named SOC-Lab was created under the domain root, with sub-OUs for Users, Groups, Workstations, and Servers. This structure keeps lab accounts, security groups, and machine objects logically separated, which mirrors how a real enterprise domain is organized.

A domain user, lab.user, was created inside the Users OU, and a security group named SOC-Analyst was created inside the Groups OU. lab.user was then added as a member of the SOC-Analyst group to simulate role-based access for a SOC team member.

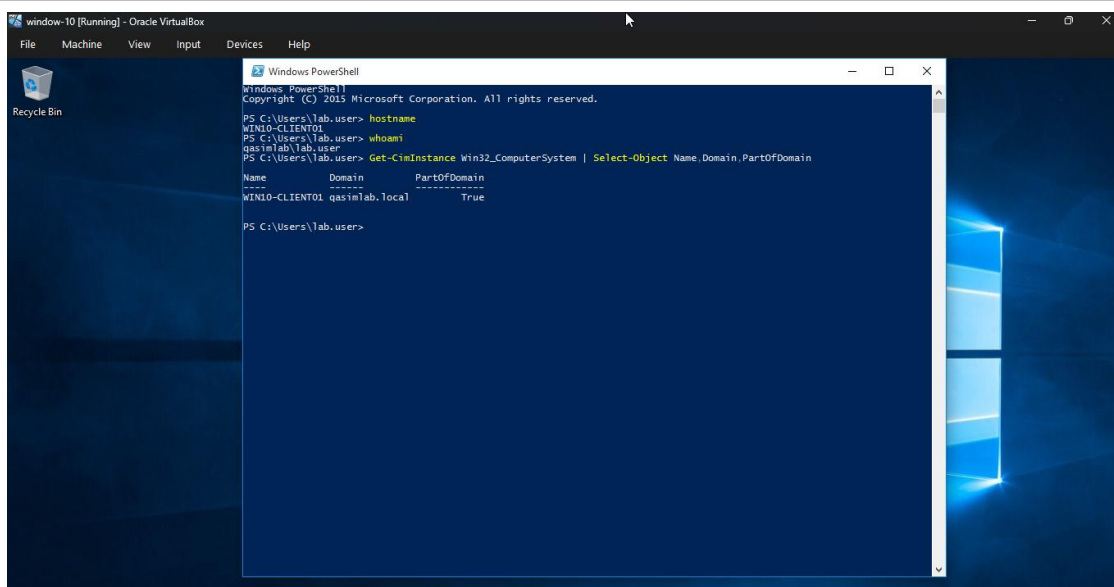


Active Directory Users and Computers: SOC-Lab OU structure and SOC-Analyst group membership showing lab.user.

4. Windows Client Domain Join

A separate Windows 10 virtual machine, WIN10-CLIENT01, was deployed and configured to use the Domain Controller (192.168.56.110) as its primary DNS server. The client was then joined to the qasimlab.local domain.

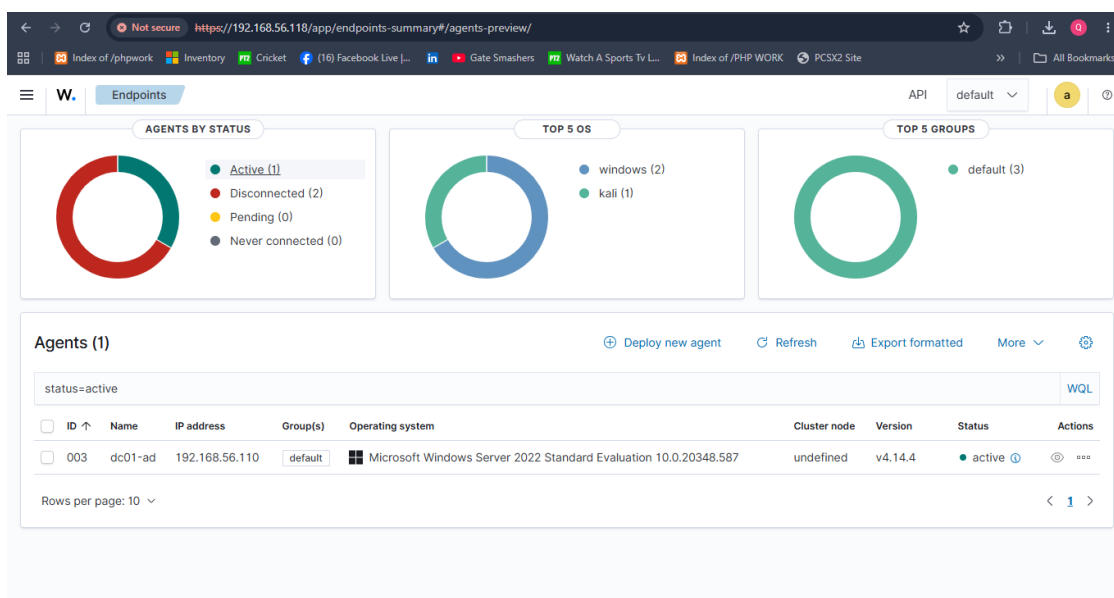
Domain membership was verified using `Get-CimInstance Win32_ComputerSystem`, which confirmed the computer name (WIN10-CLIENT01), the domain (qasimlab.local), and `PartOfDomain = True`. Sign-in was successfully performed using the domain account `qasimlab\lab.user`, confirming that authentication against the Domain Controller was working end-to-end.



WIN10-CLIENT01 verified as a domain member of qasimlab.local, signed in as qasimlab\lab.user.

5. Wazuh Agent Deployment on the Domain Controller

Wazuh agent version 4.14.4 was installed on DC01 and pointed to the Wazuh Manager at 192.168.56.118. Once installed, the Domain Controller appeared in the Wazuh Endpoints dashboard as an active agent named dc01-ad with Agent ID 003, running Microsoft Windows Server 2022 Standard Evaluation.



Wazuh Endpoints dashboard: dc01-ad (Agent ID 003) shown as Active, alongside the previous lab endpoints.

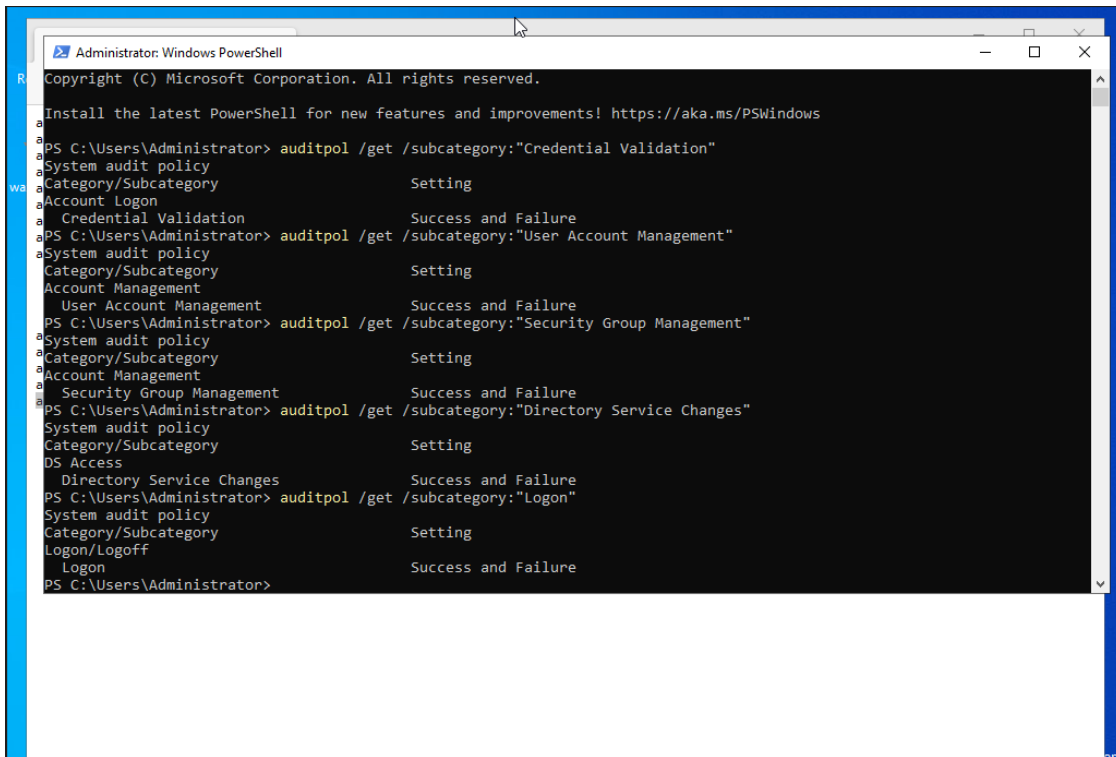
This confirmed that the Domain Controller was successfully forwarding logs to the Wazuh Manager and was ready to be monitored for Active Directory security events.

6. Advanced Windows Audit Policy Configuration

By default, Windows does not log the full set of Active Directory management events needed for security monitoring. To capture meaningful AD activity, Advanced Audit Policy was configured on DC01 for the following subcategories, with Success and Failure auditing enabled on each:

- Credential Validation
- User Account Management
- Security Group Management
- Directory Service Changes
- Logon

These subcategories were verified using `auditpol /get`, confirming that each policy was correctly applied for Success and Failure events. This ensured that account creation, group membership changes, credential validation attempts, and logon activity would all be recorded in the Windows Security event log for Wazuh to collect.



```

Administrator: Windows PowerShell
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PS C:\Users\Administrator> auditpol /get /subcategory:"Credential Validation"
System audit policy
Category/Subcategory      Setting
Account Logon
  Credential Validation    Success and Failure
PS C:\Users\Administrator> auditpol /get /subcategory:"User Account Management"
System audit policy
Category/Subcategory      Setting
Account Management
  User Account Management Success and Failure
PS C:\Users\Administrator> auditpol /get /subcategory:"Security Group Management"
System audit policy
Category/Subcategory      Setting
Account Management
  Security Group Management Success and Failure
PS C:\Users\Administrator> auditpol /get /subcategory:"Directory Service Changes"
System audit policy
Category/Subcategory      Setting
DS Access
  Directory Service Changes Success and Failure
PS C:\Users\Administrator> auditpol /get /subcategory:"Logon"
System audit policy
Category/Subcategory      Setting
Logon/Logoff
  Logon                    Success and Failure
PS C:\Users\Administrator>
  
```

auditpol /get output confirming Success and Failure auditing for Credential Validation, User Account Management, Security Group Management, Directory Service Changes, and Logon.

7. Attack Simulation: Backdoor Account Detection

To validate the complete monitoring pipeline, a realistic backdoor-account attack scenario was simulated on the domain. This technique is commonly used by attackers to establish persistence by creating a hidden privileged account, using it, and then attempting to remove evidence.

The simulated attack consisted of the following steps:

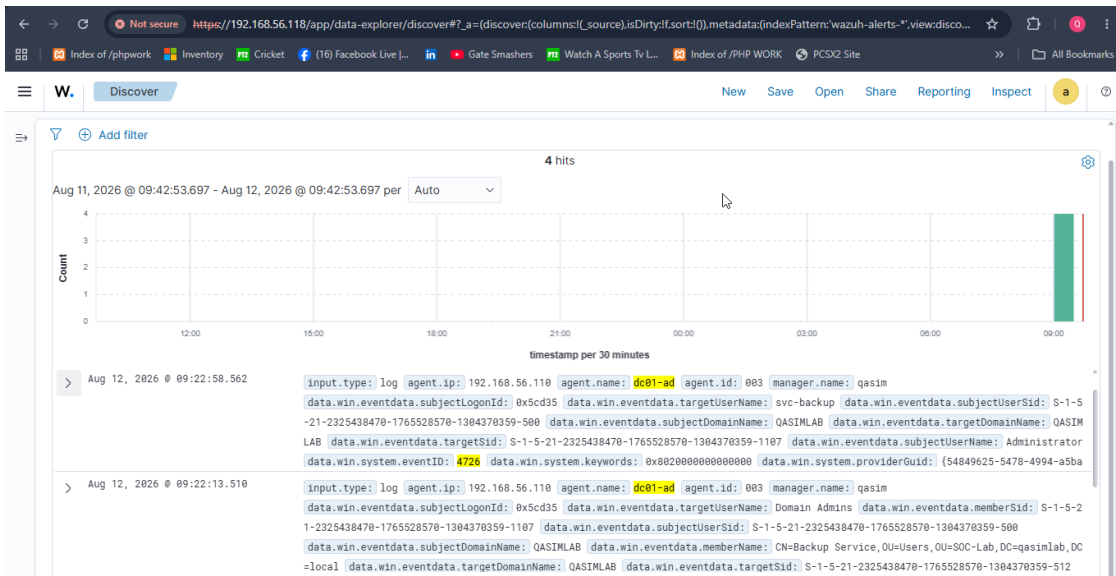
- Created a suspicious service-style account named svc-backup

- Added svc-backup to the Domain Admins group (privilege escalation)
- Removed svc-backup from the Domain Admins group
- Deleted the svc-backup account (evidence removal / cleanup)

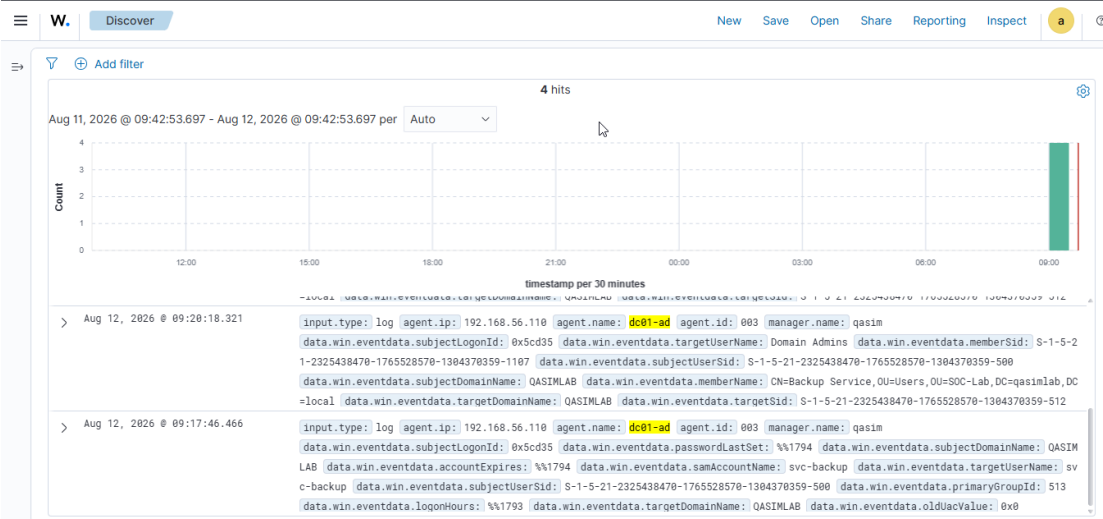
Each of these actions generated a corresponding Windows Security event, made possible by the Advanced Audit Policy configured in the previous section:

Event ID	Description
4720	A user account was created
4728	A member was added to a security-enabled global group
4729	A member was removed from a security-enabled global group
4726	A user account was deleted

Wazuh successfully collected and indexed all four events from the dc01-ad agent. The Discover view in the Wazuh dashboard showed each event with full context, including the agent name (dc01-ad), the target account (svc-backup), the affected group (Domain Admins), the initiating account (Administrator), the domain (QASIMLAB), the relevant SIDs, timestamps, and the Windows Event ID (e.g. 4726 for account deletion, 4728/4729 for group membership changes).



Wazuh Discover view: svc-backup account activity showing Event ID 4726 (account deleted) and Event ID 4728 (added to Domain Admins) on agent dc01-ad.



Wazuh Discover view: additional detail on the svc-backup account, including logon ID, domain SID, password metadata, and the Domain Admins membership change (Event ID 4729).

This confirmed that the full detection chain worked as intended: the attacker action on the Domain Controller was audited by Windows, forwarded by the Wazuh agent, and made searchable and investigable in the Wazuh dashboard.

8. Conclusion

This project extended the earlier standalone Wazuh SIEM lab into a domain-based environment, providing hands-on experience with the full lifecycle of Active Directory security monitoring. A Windows Server 2022 Domain Controller was deployed and verified, an OU/user/group structure was built, a Windows 10 client was joined to the domain, and the Domain Controller was onboarded into Wazuh as a monitored endpoint.

Advanced Audit Policy was configured to ensure that credential validation, account management, group management, directory service changes, and logon activity were all recorded. A realistic backdoor-account attack — account creation, privilege escalation into Domain Admins, privilege removal, and account deletion — was simulated and successfully detected end-to-end through Wazuh, with all four Windows Security events (4720, 4728, 4729, 4726) captured, indexed, and investigable on the dc01-ad agent.

Overall, this project builds directly on the earlier endpoint-focused Wazuh SIEM lab and demonstrates a more complete SOC workflow: from Active Directory administration and Windows security auditing, through log forwarding, to detection and investigation of a simulated identity-based attack — reinforcing practical skills in both AD administration and SIEM-based threat detection.